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STAT 4263/5263 — Statistical Inference and Time Series Modeling
Homework 1 - 2 Oct 2026

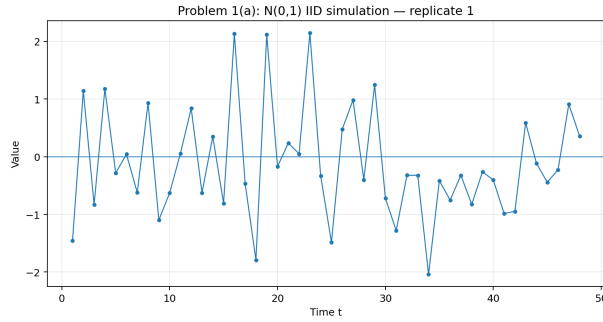
Problem 1

(a)

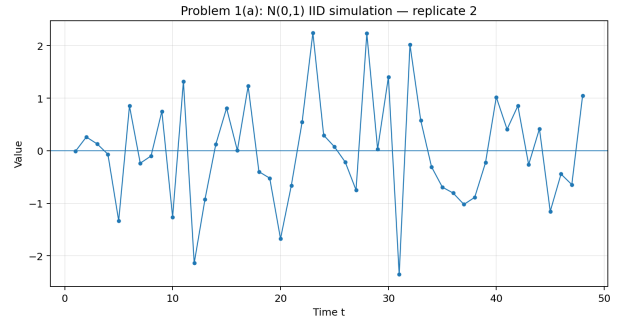
Using Python, I generated four independent realizations of length $n = 48$ from

$$X_t \stackrel{\text{iid}}{\sim} N(0, 1).$$

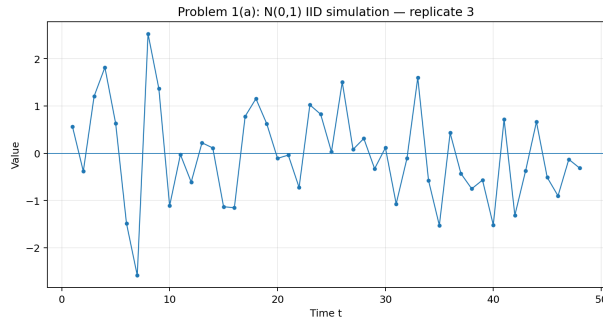
The four simulated realizations are shown below.



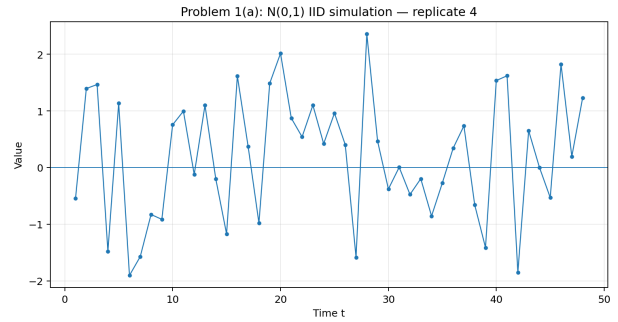
(a) Replicate 1



(b) Replicate 2



(c) Replicate 3



(d) Replicate 4

Figure 1: Four independent realizations of length 48 from $N(0, 1)$.

The individual realizations differ substantially despite being generated from the same independent process. The lag-1 sample autocorrelations for the four runs were

$$-0.2189, \quad -0.1520, \quad 0.0400, \quad -0.0754.$$

(b)

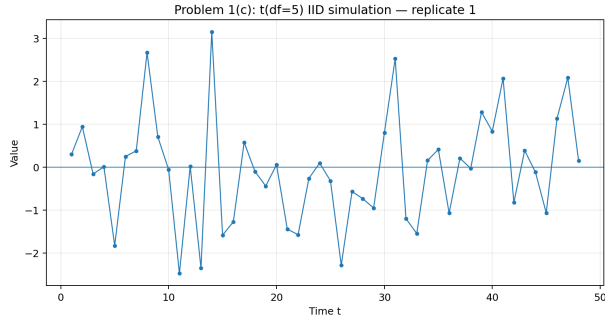
I generated four independent realizations of length 48 from

$$X_t \stackrel{\text{iid}}{\sim} \chi_2^2.$$

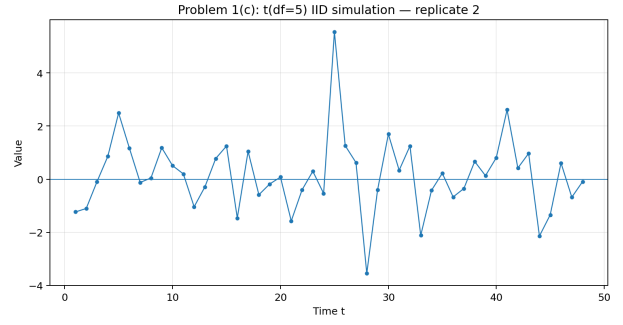
¹github.com/Nathaniel-Coulter/Timeseries-StatInf-columbia

(c)

I generated four independent realizations of length 48 from $X_t \stackrel{\text{iid}}{\sim} t_5$.



(a) Realization 1

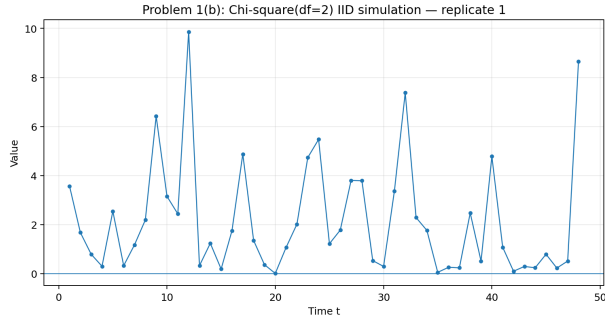


(b) Realization 2

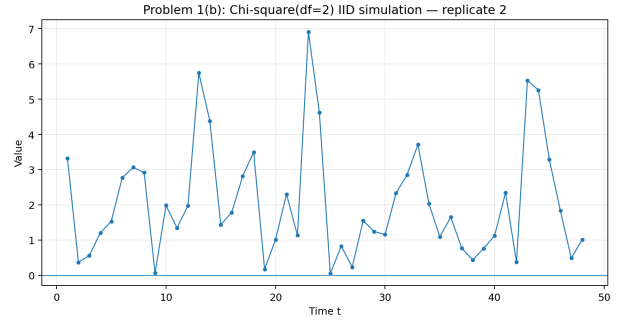
Figure 2: Two representative realizations of 48 independent t_5 observations.

The simulated paths again vary considerably across repetitions. The lag-1 sample autocorrelations were

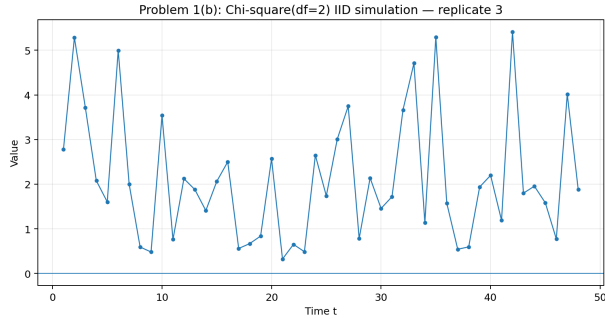
$$-0.0202, \quad 0.0948, \quad 0.2315, \quad -0.1426.$$



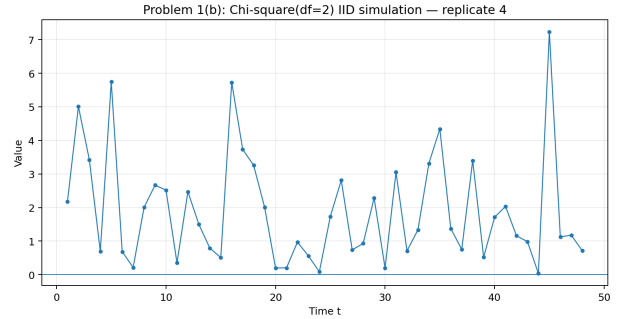
(a) Replicate 1



(b) Replicate 2



(c) Replicate 3



(d) Replicate 4

Figure 3: Four independent realizations of length 48 from χ^2_2 .

As expected for a chi-square distribution, the observations are nonnegative and right-skewed. The lag-1 sample autocorrelations across the four simulations were

$$0.0947, \quad 0.2880, \quad 0.0243, \quad -0.0489.$$

(c)

I generated four independent realizations of length 48 from $X_t \stackrel{\text{iid}}{\sim} t_5$.

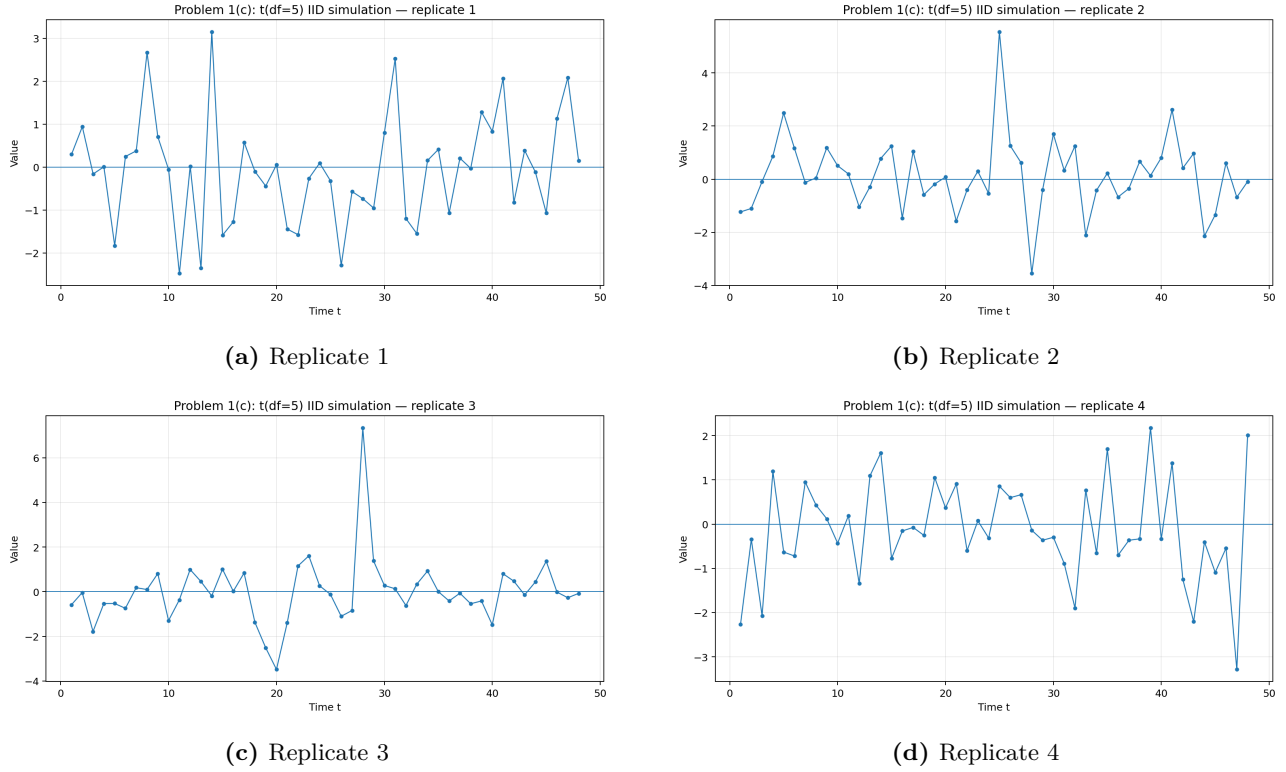


Figure 4: Four independent realizations of length 48 from a t -distribution with 5 degrees of freedom.

The simulated paths again vary considerably across repetitions. The lag-1 sample autocorrelations were

$$-0.0202, \quad 0.0948, \quad 0.2315, \quad -0.1426.$$

(d)

For $n = 48$, the approximate benchmark discussed in lecture is

$$\pm \frac{1.96}{\sqrt{n}} = \pm \frac{1.96}{\sqrt{48}} \approx \pm 0.2829.$$

The three reported lag-1 sample autocorrelations satisfy

$$|0.27| < 0.2829, \quad |-0.08| < 0.2829, \quad |0.04| < 0.2829.$$

Thus, all three values lie within the approximate bounds. In particular, the value

$$\hat{\rho}(1) = 0.27$$

for the normal series is not, by itself, sufficient evidence that the observations are dependent.

The AI response therefore overinterprets the observed sample autocorrelation. A sample autocorrelation is itself random and can be noticeably different from zero even when the underlying observations are independent.

Repeating the simulations illustrates this sampling variability. For example, in the independent chi-square simulations above, one realization produced

$$\hat{\rho}(1) = 0.2880,$$

even though the series was generated to be IID. This value is slightly outside the approximate bound of 0.2829, despite the observations being independent by construction. Different realizations of the same independent process can therefore produce substantially different sample autocorrelations.

Problem 2

Recall that a process is weakly stationary if its mean is constant and its autocovariance depends only on the lag h , not on the particular time t .

(a)

Let

$$X_t = Z_t Z_{t-2}.$$

Since Z_t and Z_{t-2} are independent with mean zero,

$$E[X_t] = E[Z_t]E[Z_{t-2}] = 0.$$

Also,

$$\begin{aligned}\gamma_X(0) &= E[X_t^2] \\ &= E[Z_t^2 Z_{t-2}^2] \\ &= E[Z_t^2]E[Z_{t-2}^2] \\ &= 1.\end{aligned}$$

For $h \neq 0$,

$$\gamma_X(h) = E[X_t X_{t+h}].$$

The only nonzero lag at which the two products share an innovation is $|h| = 2$. For example,

$$\begin{aligned}E[X_t X_{t+2}] &= E[Z_t Z_{t-2} Z_{t+2} Z_t] \\ &= E[Z_t^2 Z_{t-2} Z_{t+2}] \\ &= E[Z_t^2]E[Z_{t-2}]E[Z_{t+2}] \\ &= 0.\end{aligned}$$

Hence all nonzero-lag autocovariances vanish. Therefore,

$$\gamma_X(h) = \begin{cases} 1, & h = 0, \\ 0, & h \neq 0, \end{cases} \quad \rho_X(h) = \begin{cases} 1, & h = 0, \\ 0, & h \neq 0. \end{cases}$$

Thus,

$$\{X_t\} \text{ is weakly stationary with mean 0.}$$

(b)

For

$$X_t = a + bZ_t + cZ_{t-1},$$

the mean is

$$E[X_t] = a + bE[Z_t] + cE[Z_{t-1}] = a.$$

Thus,

$$\mu_X = a.$$

Subtracting the mean,

$$X_t - a = bZ_t + cZ_{t-1},$$

so

$$\begin{aligned}\gamma_X(0) &= \text{Var}(bZ_t + cZ_{t-1}) \\ &= b^2 \text{Var}(Z_t) + c^2 \text{Var}(Z_{t-1}) \\ &= b^2 + c^2.\end{aligned}$$

At lag one,

$$\begin{aligned}\gamma_X(1) &= E[(X_t - a)(X_{t+1} - a)] \\ &= E[(bZ_t + cZ_{t-1})(bZ_{t+1} + cZ_t)] \\ &= bc E[Z_t^2] \\ &= bc,\end{aligned}$$

since all other cross-products have expectation zero.

By symmetry,

$$\gamma_X(-1) = bc.$$

For $|h| > 1$, the variables contain no common innovations, so

$$\gamma_X(h) = 0.$$

Therefore,

$$\gamma_X(h) = \begin{cases} b^2 + c^2, & h = 0, \\ bc, & |h| = 1, \\ 0, & |h| > 1. \end{cases}$$

Provided $b^2 + c^2 > 0$,

$$\rho_X(h) = \begin{cases} 1, & h = 0, \\ \frac{bc}{b^2 + c^2}, & |h| = 1, \\ 0, & |h| > 1. \end{cases}$$

Hence,

$$\{X_t\} \text{ is weakly stationary.}$$

(c)

For

$$X_t = Z_t - Z_{t-3},$$

we have

$$E[X_t] = 0.$$

The variance is

$$\begin{aligned}\gamma_X(0) &= \text{Var}(Z_t - Z_{t-3}) \\ &= \text{Var}(Z_t) + \text{Var}(Z_{t-3}) \\ &= 2.\end{aligned}$$

The only nonzero covariance occurs when the two variables share an innovation, which happens at lag 3. Since

$$X_{t+3} = Z_{t+3} - Z_t,$$

we obtain

$$\begin{aligned}\gamma_X(3) &= E[(Z_t - Z_{t-3})(Z_{t+3} - Z_t)] \\ &= -E[Z_t^2] \\ &= -1.\end{aligned}$$

By symmetry,

$$\gamma_X(-3) = -1.$$

Therefore,

$$\gamma_X(h) = \begin{cases} 2, & h = 0, \\ -1, & |h| = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Dividing by $\gamma_X(0) = 2$,

$$\rho_X(h) = \begin{cases} 1, & h = 0, \\ -\frac{1}{2}, & |h| = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$\boxed{\{X_t\} \text{ is weakly stationary with mean 0.}}$$

(d)

Consider

$$X_t = Z_t \cos(ct) + Z_{t-1} \sin(ct).$$

Since both innovations have mean zero,

$$E[X_t] = 0.$$

The variance is

$$\begin{aligned}\text{Var}(X_t) &= \cos^2(ct) \text{Var}(Z_t) + \sin^2(ct) \text{Var}(Z_{t-1}) \\ &= \cos^2(ct) + \sin^2(ct) \\ &= 1.\end{aligned}$$

However,

$$X_{t+1} = Z_{t+1} \cos(c(t+1)) + Z_t \sin(c(t+1)),$$

so the only common innovation between X_t and X_{t+1} is Z_t . Hence,

$$\begin{aligned}\text{Cov}(X_t, X_{t+1}) &= \cos(ct) \sin(c(t+1)) \\ &= \frac{1}{2} [\sin(2ct + c) + \sin(c)].\end{aligned}$$

This generally depends on t , rather than only on the lag. Therefore,

$$\boxed{\{X_t\} \text{ is generally not weakly stationary.}}$$

There is a special case. If

$$c \in \pi\mathbb{Z},$$

then $\sin(ct) = 0$ for integer t , and

$$X_t = \cos(ct)Z_t = (\pm 1)Z_t.$$

Because the standard normal distribution is symmetric, this is again an i.i.d. $N(0, 1)$ sequence. Thus,

$$\boxed{\text{when } c \in \pi\mathbb{Z}, \text{ the process is stationary.}}$$

(e)

The AI response is incorrect. Zero autocorrelation implies that X_t and X_{t+2} are uncorrelated, but it does not imply that they are independent.

From part (a),

$$X_t = Z_t Z_{t-2}, \quad X_{t+2} = Z_{t+2} Z_t.$$

The two random variables share the innovation Z_t . Squaring and multiplying gives

$$X_t^2 X_{t+2}^2 = Z_t^4 Z_{t-2}^2 Z_{t+2}^2.$$

Since Z_t , Z_{t-2} , and Z_{t+2} are independent,

$$\begin{aligned} E[X_t^2 X_{t+2}^2] &= E[Z_t^4] E[Z_{t-2}^2] E[Z_{t+2}^2] \\ &= 3 \cdot 1 \cdot 1 \\ &= 3. \end{aligned}$$

On the other hand,

$$\begin{aligned} E[X_t^2] &= E[Z_t^2] E[Z_{t-2}^2] \\ &= 1, \end{aligned}$$

and similarly,

$$E[X_{t+2}^2] = 1.$$

Therefore,

$$E[X_t^2] E[X_{t+2}^2] = 1.$$

Hence,

$$E[X_t^2 X_{t+2}^2] = 3 \neq 1 = E[X_t^2] E[X_{t+2}^2].$$

If X_t and X_{t+2} were independent, then functions of them, including X_t^2 and X_{t+2}^2 , would also be independent, and the expectations above would factor. They do not.

Therefore,

$$X_t \text{ and } X_{t+2} \text{ are dependent even though they are uncorrelated.}$$

Consequently,

$$\{X_t\} \text{ in part (a) is not i.i.d.}$$

Thus the AI response confuses zero correlation with independence.

Summary

Process	Weakly stationary?
(a) $Z_t Z_{t-2}$	Yes
(b) $a + bZ_t + cZ_{t-1}$	Yes
(c) $Z_t - Z_{t-3}$	Yes
(d) $Z_t \cos(ct) + Z_{t-1} \sin(ct)$	Generally no

For part (d), the special case $c \in \pi\mathbb{Z}$ is stationary.

$$\text{independence} \implies \text{uncorrelated}, \quad \text{but} \quad \text{uncorrelated} \not\Rightarrow \text{independence.}$$

Problem 3

Let X be a random variable with

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2,$$

and define

$$Y_t = X \quad \text{for all } t.$$

(a)

To show strict stationarity, consider any collection of time points $t_1, \dots, t_k$ and any shift h . Since $Y_t = X$ for every t ,

$$(Y_{t_1}, \dots, Y_{t_k}) = (X, \dots, X),$$

while

$$(Y_{t_1+h}, \dots, Y_{t_k+h}) = (X, \dots, X).$$

Thus the two random vectors are identical and therefore have the same joint distribution. Hence $\{Y_t\}$ is strictly stationary.

For weak stationarity,

$$E[Y_t] = E[X] = \mu,$$

which does not depend on t . Moreover, for any lag h ,

$$\begin{aligned} \text{Cov}(Y_t, Y_{t+h}) &= \text{Cov}(X, X) \\ &= \text{Var}(X) \\ &= \sigma^2. \end{aligned}$$

The mean is constant and the autocovariance depends only on the lag (in fact, it is the same for every lag). Therefore,

$\{Y_t\}$ is both strictly and weakly stationary.

(b)

Since $Y_t = Y_{t+h} = X$ for every t and h ,

$\gamma_Y(h) = \text{Cov}(Y_t, Y_{t+h}) = \sigma^2, \quad h \in \mathbb{Z}.$

Thus the autocovariance is constant at every lag. If $\sigma^2 > 0$, the corresponding autocorrelation is

$$\rho_Y(h) = 1 \quad \text{for every } h.$$

(c)

For any fixed realization ω ,

$$Y_t(\omega) = X(\omega) \quad \text{for every } t.$$

Therefore every sample path is a horizontal line at the realized value of X .

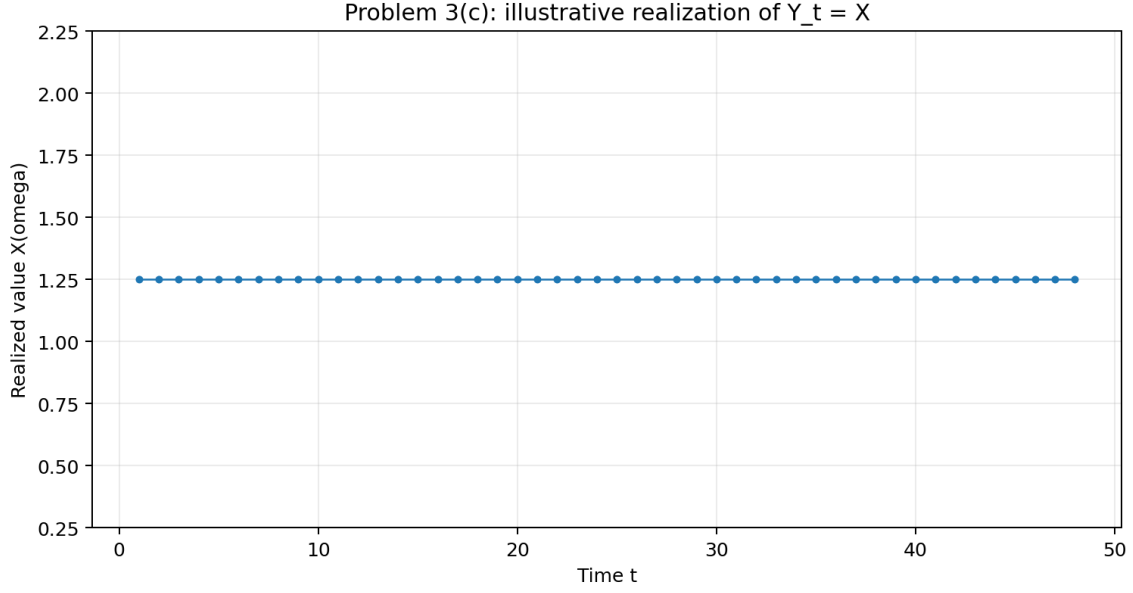


Figure 5: An illustrative realization of $Y_t = X$. Here the arbitrary realized value is $X(\omega) = 1.25$; any realization would be a horizontal line at its corresponding value of X .

(d)

The AI argument is incorrect. Although $Y_1, \dots, Y_n$ are recorded at n different times, they are not independent observations.

Using the autocovariance function from part (b),

$$\begin{aligned}
 \text{Var}(\bar{Y}_n) &= \text{Var}\left(\frac{1}{n} \sum_{t=1}^n Y_t\right) \\
 &= \frac{1}{n^2} \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(Y_s, Y_t) \\
 &= \frac{1}{n^2} \sum_{s=1}^n \sum_{t=1}^n \sigma^2 \\
 &= \frac{n^2 \sigma^2}{n^2} \\
 &= \boxed{\sigma^2}.
 \end{aligned}$$

Thus

$$\boxed{\text{Var}(\bar{Y}_n) = \sigma^2}$$

for every n . The variance does not decrease as the number of observed time points increases.



Simpler takeaway, courtesy of ChatGPT. Because every observation is exactly the same random variable X ,

$$\bar{Y}_n = \frac{1}{n} \underbrace{(X + X + \dots + X)}_{n \text{ copies}} = \frac{nX}{n} = \boxed{X}.$$

Therefore,

$$\text{Var}(\bar{Y}_n) = \text{Var}(X) = \sigma^2.$$

This also gives the intuition directly. After observing $Y_1 = X$, every later observation simply repeats the same realized value:

$$Y_2 = Y_1, \quad Y_3 = Y_1, \quad \dots$$

Thus n recorded values do not provide n independent pieces of information about μ . Observing the process for a longer period adds no new information about the distributional mean, so the sample mean does not become more precise as n increases.

Problem 4

Consider

$$X_t = \sin(2\pi Ut), \quad t = 1, 2, \dots,$$

where

$$U \sim \text{Unif}(0, 1).$$

(a)

For every positive integer t ,

$$\begin{aligned} E[X_t] &= \int_0^1 \sin(2\pi tu) \, du \\ &= 0. \end{aligned}$$

Thus the mean is constant:

$$\boxed{E[X_t] = 0.}$$

Since the mean is zero,

$$\text{Cov}(X_t, X_s) = E[X_t X_s].$$

Using

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)],$$

we obtain

$$\begin{aligned} E[X_t X_s] &= \int_0^1 \sin(2\pi tu) \sin(2\pi su) \, du \\ &= \frac{1}{2} \int_0^1 [\cos(2\pi(t - s)u) - \cos(2\pi(t + s)u)] \, du. \end{aligned}$$

If $t \neq s$, both cosine terms integrate to zero, so

$$E[X_t X_s] = 0.$$

If $t = s$,

$$\begin{aligned} E[X_t^2] &= \int_0^1 \sin^2(2\pi tu) \, du \\ &= \frac{1}{2}. \end{aligned}$$

Therefore, in terms of the lag h ,

$$\boxed{\gamma_X(h) = \begin{cases} \frac{1}{2}, & h = 0, \\ 0, & h \neq 0. \end{cases}}$$

Hence the autocorrelation function is

$$\boxed{\rho_X(h) = \begin{cases} 1, & h = 0, \\ 0, & h \neq 0. \end{cases}}$$

(b)

From part (a),

$$E[X_t] = 0$$

for every t , and

$$\gamma_X(h) = \begin{cases} \frac{1}{2}, & h = 0, \\ 0, & h \neq 0, \end{cases}$$

depends only on the lag h , not on the particular time t .

Therefore,

$$\boxed{\{X_t\} \text{ is weakly stationary.}}$$

(c)

Let

$$a = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}.$$

For X_1 ,

$$X_1 > a \iff U \in \left(\frac{1}{6}, \frac{1}{3}\right).$$

For X_2 ,

$$X_2 > a \iff U \in \left(\frac{1}{12}, \frac{1}{6}\right) \cup \left(\frac{7}{12}, \frac{2}{3}\right).$$

These sets are disjoint. Hence,

$$\boxed{P(X_1 > a, X_2 > a) = 0.}$$

For X_3 ,

$$X_3 > a \iff U \in \left(\frac{1}{18}, \frac{1}{9}\right) \cup \left(\frac{7}{18}, \frac{4}{9}\right) \cup \left(\frac{13}{18}, \frac{7}{9}\right).$$

The only overlap of the events $\{X_2 > a\}$ and $\{X_3 > a\}$ is

$$U \in \left(\frac{1}{12}, \frac{1}{9}\right).$$

Because U is uniform on $(0, 1)$,

$$\begin{aligned} P(X_2 > a, X_3 > a) &= \frac{1}{9} - \frac{1}{12} \\ &= \frac{4}{36} - \frac{3}{36} \\ &= \frac{1}{36}. \end{aligned}$$

Thus,

$$\boxed{P(X_2 > a, X_3 > a) = \frac{1}{36}.}$$

Therefore,

$$P(X_1 > a, X_2 > a) \neq P(X_2 > a, X_3 > a).$$

The joint distribution of consecutive observations therefore changes under a shift in time. Hence,

$$\boxed{\{X_t\} \text{ is not strictly stationary.}}$$

(d)

The variance calculation in the AI response is justified, but the application of the i.i.d. central limit theorem is not.

From part (a),

$$\gamma_X(0) = \frac{1}{2}, \quad \gamma_X(h) = 0 \quad \text{for } h \neq 0.$$

Therefore,

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum_{t=1}^n X_t\right) \\ &= \frac{1}{n^2} \sum_{s=1}^n \sum_{t=1}^n \text{Cov}(X_s, X_t) \\ &= \frac{1}{n^2} \left(n \cdot \frac{1}{2}\right) \\ &= \boxed{\frac{1}{2n}}. \end{aligned}$$

Thus the AI is correct that

$$\boxed{\text{Var}(\bar{X}_n) = \frac{1}{2n}}.$$

This calculation does not require independence; the autocovariance function alone is sufficient.

However, zero autocorrelation at nonzero lags implies only that the observations are uncorrelated. It does not imply that they are independent. Indeed, part (c) shows directly that the joint distributions exhibit dependence and are not invariant under time shifts.

Consequently, the usual i.i.d. central limit theorem cannot be invoked merely from

$$\rho_X(h) = 0, \quad h \neq 0.$$

Thus the conclusion

$$\sqrt{n} \bar{X}_n \approx N\left(0, \frac{1}{2}\right)$$

does not automatically follow from the zero autocorrelations.

The two main distinctions illustrated by this problem are

$$\boxed{\text{weak stationarity} \not\Rightarrow \text{strict stationarity}}$$

and

$$\boxed{\text{uncorrelated} \not\Rightarrow \text{independent}}.$$

Problem 5

Let

$$Z_t = X_t + Y_t,$$

where $\{X_t\}$ and $\{Y_t\}$ are weakly stationary and mutually uncorrelated:

$$\text{Cov}(X_s, Y_t) = 0 \quad \text{for all } s, t.$$

(a)

Let

$$E[X_t] = \mu_X, \quad E[Y_t] = \mu_Y.$$

Then

$$E[Z_t] = E[X_t] + E[Y_t] = \mu_X + \mu_Y,$$

which is constant in t .

For any lag h ,

$$\begin{aligned} \gamma_Z(h) &= \text{Cov}(Z_t, Z_{t+h}) \\ &= \text{Cov}(X_t + Y_t, X_{t+h} + Y_{t+h}) \\ &= \text{Cov}(X_t, X_{t+h}) + \text{Cov}(X_t, Y_{t+h}) \\ &\quad + \text{Cov}(Y_t, X_{t+h}) + \text{Cov}(Y_t, Y_{t+h}). \end{aligned}$$

Since X_s and Y_t are uncorrelated for every s and t , the two cross-covariance terms vanish. Therefore,

$$\boxed{\gamma_Z(h) = \gamma_X(h) + \gamma_Y(h).}$$

Because both $\gamma_X(h)$ and $\gamma_Y(h)$ depend only on the lag h , so does $\gamma_Z(h)$. Hence,

$$\boxed{\{Z_t\} \text{ is weakly stationary.}}$$

(b)

The AI response is incorrect. Equal and opposite correlations do not necessarily cancel because the two processes may have different variance scales.

Recall that

$$\gamma(1) = \rho(1)\gamma(0) = \rho(1) \text{Var}(X_t).$$

Thus,

$$\gamma_X(1) = \rho_X(1) \text{Var}(X_t) = 0.8(1) = 0.8,$$

while

$$\gamma_Y(1) = \rho_Y(1) \text{Var}(Y_t) = (-0.8)(4) = -3.2.$$

From part (a),

$$\boxed{\gamma_Z(0) = \gamma_X(0) + \gamma_Y(0) = 1 + 4 = 5.}$$

Similarly,

$$\boxed{\gamma_Z(1) = \gamma_X(1) + \gamma_Y(1) = 0.8 - 3.2 = -2.4.}$$

Therefore,

$$\boxed{\rho_Z(1) = \frac{\gamma_Z(1)}{\gamma_Z(0)} = \frac{-2.4}{5} = -0.48.}$$

Hence the lag-one effects do not cancel, and the AI conclusion

$$\rho_Z(1) = 0$$

is incorrect.

More generally, suppose

$$\rho_X(1) = r, \quad \rho_Y(1) = -r, \quad r \neq 0.$$

Let

$$\sigma_X^2 = \text{Var}(X_t), \quad \sigma_Y^2 = \text{Var}(Y_t).$$

Then

$$\gamma_X(1) = r\sigma_X^2, \quad \gamma_Y(1) = -r\sigma_Y^2,$$

so

$$\begin{aligned} \gamma_Z(1) &= r\sigma_X^2 - r\sigma_Y^2 \\ &= r(\sigma_X^2 - \sigma_Y^2). \end{aligned}$$

Since $r \neq 0$,

$$\gamma_Z(1) = 0 \iff \sigma_X^2 = \sigma_Y^2.$$

Therefore,

$$\boxed{\text{Var}(X_t) = \text{Var}(Y_t)}$$

is the additional condition required for the two lag-one effects to cancel.

Equivalently, the key relationship is

$$\boxed{\gamma(1) = \rho(1) \text{Var}(X_t)}.$$

Thus equal and opposite correlations cancel only when they are measured on the same variance scale.

Problem 6

The first retail series in `retail.xlsx` contains monthly New South Wales retail turnover data². The observations run from April 1982 through July 2022.

(a)

The series plotted sequentially against time is shown below.

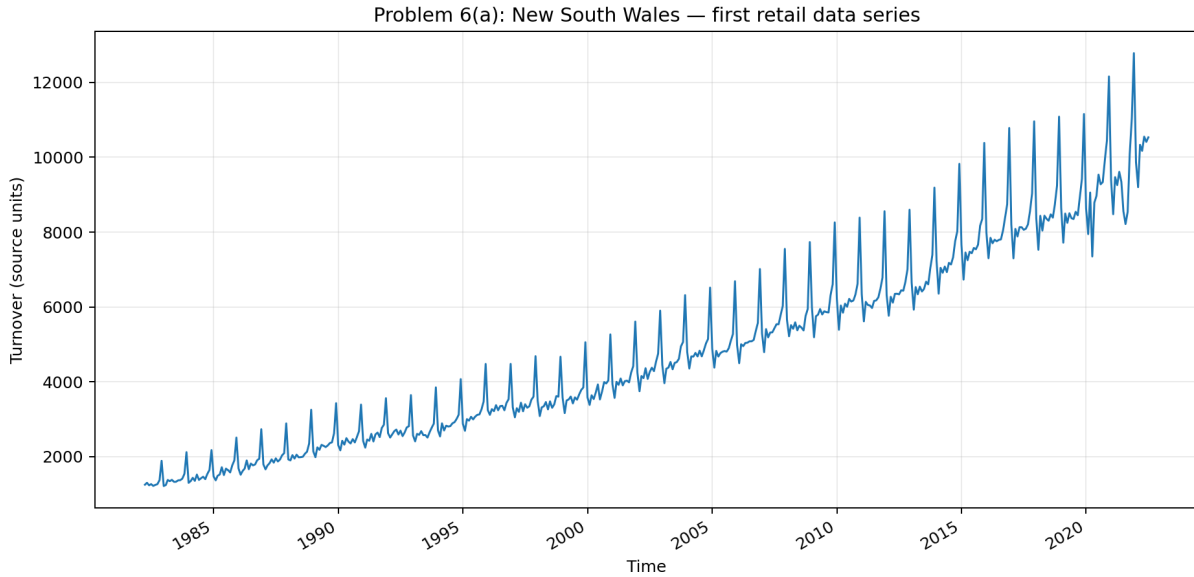


Figure 6: Monthly New South Wales retail turnover for the first series in `retail.xlsx`.

²The assignment describes the first series as supermarket and grocery-store sales. The CourseWorks workbook itself labels the first numerical series as “Turnover; New South Wales; Total (Industry)” (Series ID A3348645F). Following the assignment’s instruction to use the first column, I analyze that first numerical series.

The series does not appear to be a realization of a weakly stationary process. There is a pronounced upward trend over the sample period, so the mean level is clearly not approximately constant through time. For example, the mean of the first 12 observations is approximately

$$1322.3,$$

whereas the mean of the final 12 observations is approximately

$$10145.0.$$

The plot also shows strong annual seasonality, with recurring year-end peaks. In addition, the absolute size of the seasonal fluctuations appears to increase as the overall level of the series rises.

Thus the changing level, recurring seasonal pattern, and increasing magnitude of the fluctuations all provide evidence against weak stationarity.

(b)

The seasonal pattern can be displayed in both linear and cyclical forms.

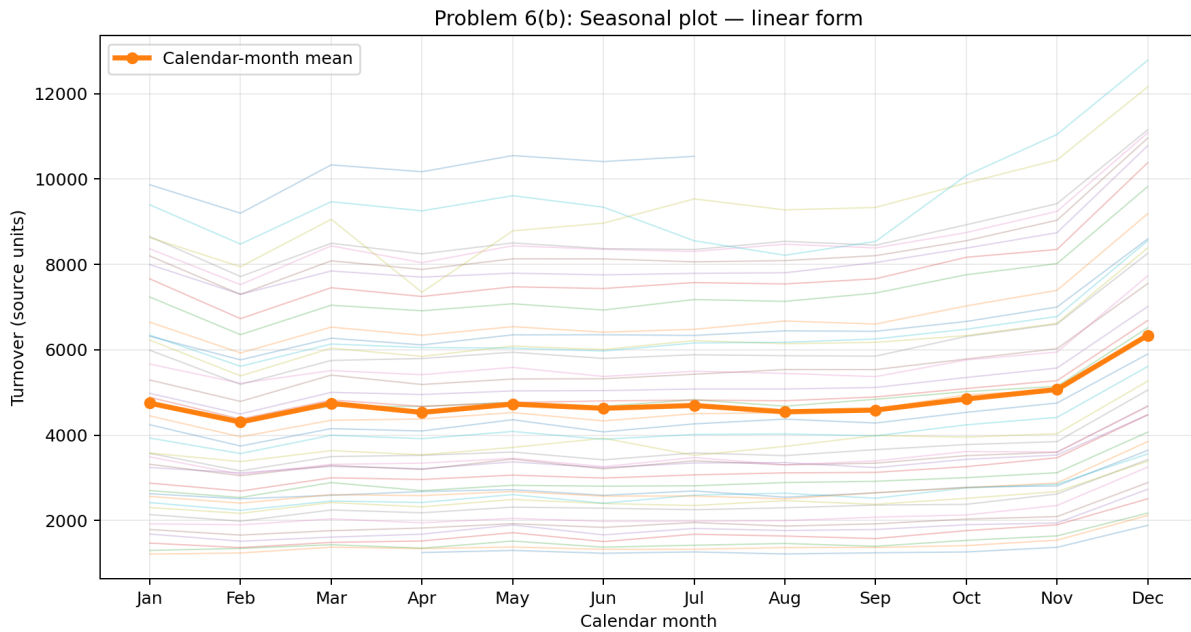


Figure 7: Linear seasonal plot of New South Wales retail turnover. Individual lines represent yearly seasonal cycles, while the emphasized line shows the calendar-month mean.

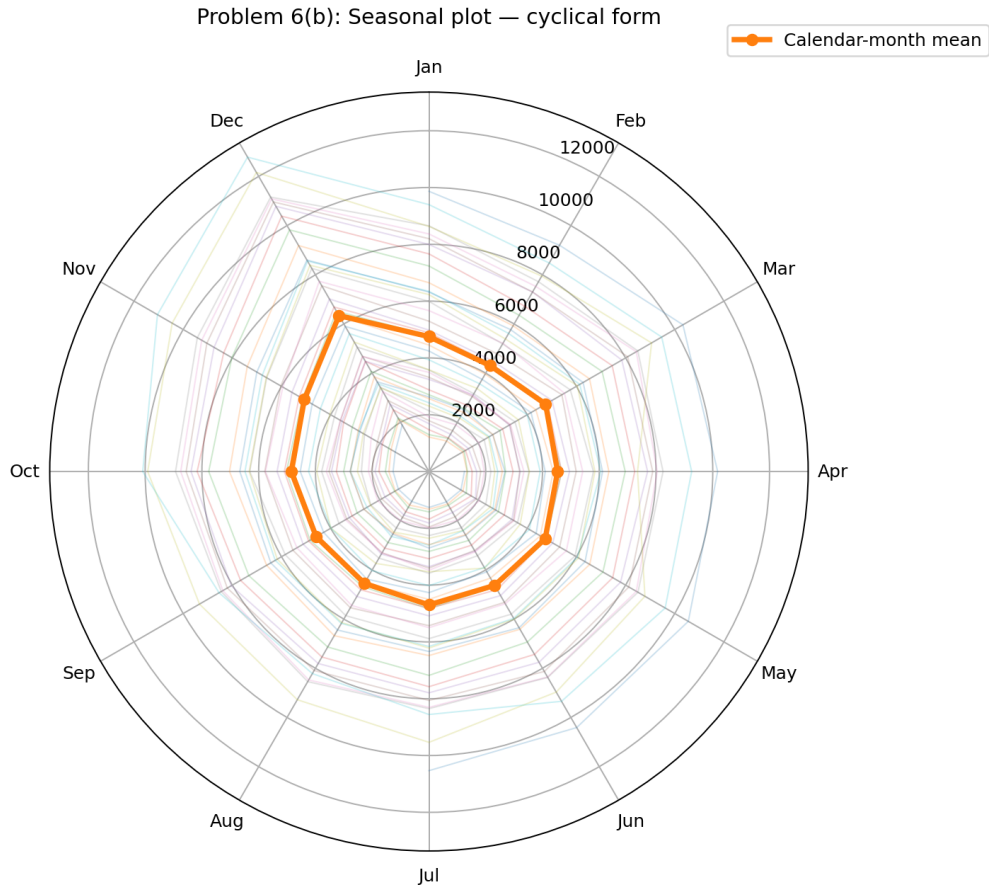


Figure 8: Cyclical seasonal plot of New South Wales retail turnover.

Both plots show a clear recurring annual pattern. In particular, December has the highest average sales in the raw data. Its calendar-month mean is approximately

$$6340.3.$$

For comparison, the November mean is approximately 5070.7, so the December increase is especially pronounced.

Therefore,

$$\text{December has the highest average sales.}$$

These are raw calendar-month averages; the series has not first been detrended.

(c)

Using the sample autocovariance definition given in the problem,

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (x_{t+h} - \bar{x}_n)(x_t - \bar{x}_n),$$

the sample autocorrelation function is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

The sample ACF through lag 60 months is shown below.

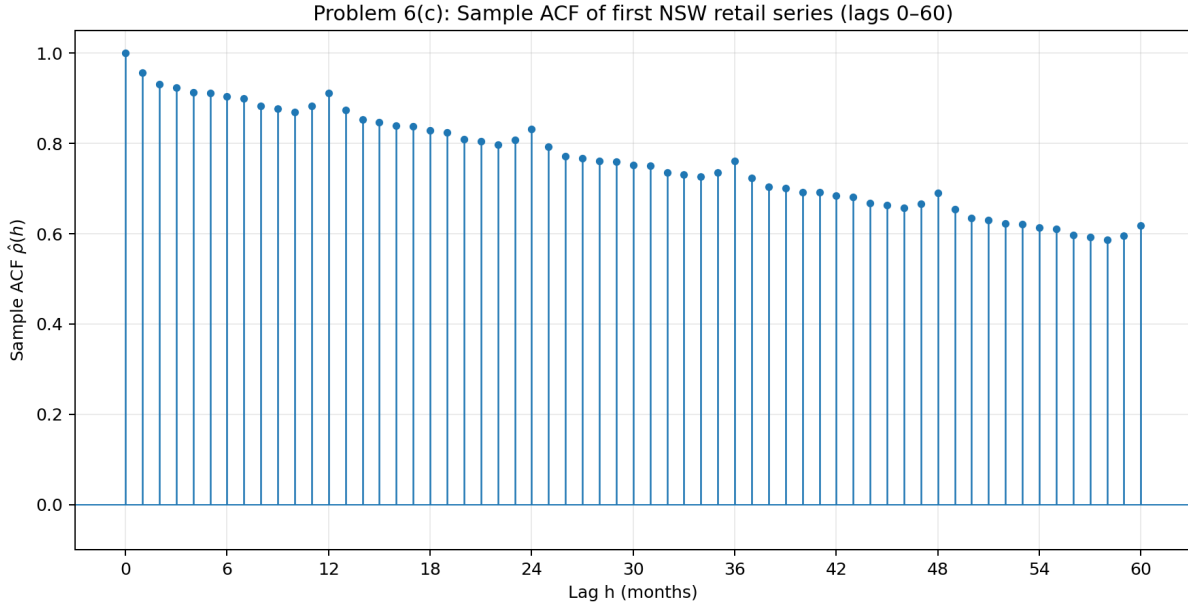


Figure 9: Sample autocorrelation function for the New South Wales retail series at lags 0 through 60 months.

The sample ACF is large and positive across many lags and decays only very slowly. For example,

$$\hat{\rho}(1) \approx 0.9563, \quad \hat{\rho}(12) \approx 0.9121.$$

There are also noticeable increases around seasonal lags that are multiples of 12 months:

$$\begin{aligned} \hat{\rho}(12) &\approx 0.9121, \\ \hat{\rho}(24) &\approx 0.8326, \\ \hat{\rho}(36) &\approx 0.7610, \\ \hat{\rho}(48) &\approx 0.6903, \\ \hat{\rho}(60) &\approx 0.6183. \end{aligned}$$

Thus the main features are a very high short-lag autocorrelation, slow decay as the lag increases, and recurring local increases at approximately yearly intervals.

These features are consistent with the strong upward trend and annual seasonality visible in the original series. At this stage, however, the large sample autocorrelations should be interpreted descriptively rather than immediately as evidence of a stable stationary dependence structure.

(d)

The AI response is only partly correct.

It is reasonable to describe the observed sample ACF as large and positive over many lags, including around lag 12. For example,

$$\hat{\rho}(1) \approx 0.9563, \quad \hat{\rho}(12) \approx 0.9121.$$

However, interpreting these values as evidence of a stable persistence structure is not justified from the raw series alone.

 **ChatGPT suggestion.** The preceding sentence could be stated more sharply because the observed large autocorrelations themselves are not in dispute; the problem is the interpretation placed on them. A more precise version would be:

“However, the further claim that these large correlations necessarily represent a stable persistence structure is not justified from the raw series alone.”

The distinction is useful because the sample ACF can be described accurately without assuming that it estimates a stable stationary dependence structure.

In particular, calculating a sample ACF does *not* remove trend or seasonality. A strong trend can produce large positive sample autocorrelations because observations that are close in time tend to lie at similar levels of the trend. Likewise, recurring seasonality can generate large autocorrelations at seasonal lags such as 12, 24, 36, . . .

This is important because, for a weakly stationary process, the mean is constant and the autocovariance

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$

depends only on the lag h , not on the particular time t .

The original retail series clearly has a changing level and strong annual seasonality. Therefore, its raw sample ACF should be viewed primarily as a descriptive feature of this observed series, rather than automatically as an estimate of a stable population ACF.

Thus,

a large raw sample ACF does not by itself establish stationary persistence.

(e)

The proposed forecasting procedure should be evaluated using one simple principle:

When predicting x_t , could this information actually have been known when the forecast was made?

1. Using the last 12 months as a test set: appropriate.

A genuine forecasting evaluation should preserve the chronological ordering of the observations. Using the final 12 months as a holdout set ensures that the model is trained only on data that occurred earlier in time.

This is preferable to a random train/test split, which could allow future observations to influence the fitted model. The final 12 months therefore form a valid test set, although 12 observations alone provide a relatively small number of forecast errors.

 **ChatGPT technical refinement.** The sentence above is directionally correct, but it can be framed more precisely for a time-series setting. The specific problem with a random split is that it may place observations occurring later in calendar time in the training set while evaluating predictions for observations that occurred earlier.

A more precise version is:

“This is preferable to a random train/test split, which can place observations occurring later in calendar time in the training set while evaluating predictions for earlier observations.”

The same time-ordering principle also applies to preprocessing: if standardization is used, its parameters should be estimated using only information available in the training period.

2. Standardizing the complete series before the train/test split: inappropriate.

If the mean and standard deviation are calculated from the complete series, then observations from the future test period influence the transformation applied to the training data. This is a form of *data leakage*.

If standardization is used, the mean and standard deviation should be estimated using only the training observations. The resulting training-set transformation can then be applied unchanged to later validation and test observations.

3. Using x_{t-12} as a predictor: appropriate.

When forecasting x_t , the observation x_{t-12} occurred one year earlier and is already known. It is therefore a legitimate predictor.

It is also particularly natural for these data because the seasonal plots show a strong annual pattern.

The fact that x_{t-12} may be strongly related to x_t is not a problem; the important point is that its value would actually have been available when the forecast was made.

4. Using x_{t+1} as a predictor: inappropriate.

The value x_{t+1} occurs after the target x_t . It would not be known when forecasting time t , so using it introduces direct look-ahead bias.

Even if x_{t+1} is strongly associated with x_t in the historical data, that relationship is not usable in a genuine forecast because it requires information from the future.

5. Using calendar month as a predictor: appropriate.

The calendar month of the forecast date is known in advance. Therefore, it introduces no future information and is a valid predictor.

It is also useful here because the seasonal plots show a recurring calendar-month pattern, with especially high sales near the end of the year.

A better evaluation procedure would preserve time ordering throughout the entire modeling process.

I would keep the final 12 months completely untouched as the final test set and use only the earlier observations for model development. Within those earlier observations, model selection and tuning could be performed using a rolling-origin or expanding-window evaluation.

At each forecast origin:

1. fit any preprocessing steps using only data available up to that time;
2. train the forecasting model using only past observations;
3. construct predictors only from information that would have been known at the forecast date; and
4. evaluate the resulting forecast on the next observation, or on the same forecast horizon that would ultimately be used in practice.

After selecting the model, it could be refit using all observations available before the final 12-month test period and then evaluated on that untouched holdout set using an appropriate forecast-error measure.

Therefore, the proposed choices can be summarized as

Proposed choice	Appropriate?
Last 12 months as test set	Yes
Standardize using complete series	No
x_{t-12} as predictor	Yes
x_{t+1} as predictor	No
Calendar month as predictor	Yes

The central distinction is therefore

predictive information is usable only if it would have been available at the forecast origin.

Computational Appendix

Problem 1: Simulation Code

The simulations and plots in Problem 1 were generated in Python. The relevant computational steps were:

```
rng = np.random.default_rng(SEED)
N = 48

x_normal = rng.normal(0.0, 1.0, size=N)
x_chi2    = rng.chisquare(df=2, size=N)
x_t5      = rng.standard_t(df=5, size=N)

def sample_acf(x, lag=1):
    x = np.asarray(x, dtype=float)
    xbar = x.mean()
    centered = x - xbar
    return (
        np.dot(centered[lag:], centered[:-lag])
        / np.dot(centered, centered)
    )
```

Problem 6: Retail-Series Analysis

The first numerical series in `retail.xlsx` was extracted together with its monthly date column. The sample ACF was computed directly from the definition given in the assignment:

```
def sample_acf_exact(values, max_lag):
    x = np.asarray(values, dtype=float)
    n = len(x)
    xbar = x.mean()
    centered = x - xbar

    gamma = np.empty(max_lag + 1)

    for h in range(max_lag + 1):
        gamma[h] = (
            np.dot(centered[h:], centered[:n-h]) / n
        )

    rho = gamma / gamma[0]
    return gamma, rho
```

Calendar-month averages used in the seasonal plots were computed by grouping the observations by month, and the sequential, seasonal, and ACF figures were produced with Matplotlib.

 **ChatGPT computational assistance.** ChatGPT assisted in designing and reviewing the Python scripts used for Problems 1 and 6, including the simulation workflow, reproducible output structure, direct implementation of the sample ACF formula, data validation, and plotting code. I interpreted the resulting figures and numerical output in the written solutions above.